👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

**UNIT 10 · STANDARD MPP.7**

# Math is Beauty

Lesson 10-2 · Teacher Copy

**Name:** Type your name **Date:** Today's date **Class:** Class period

# My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

**Content Objective: I can use bilateral symmetry to describe structure in nature and in human creations.**

**Language Objective: I can describe a symmetric design using the words mirror image, line of symmetry, and balance.**

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

symmetric | mirror image | bilateral symmetry | line of symmetry | balance

☝️ *Tap any word to see its meaning and a picture.*

**1** Having two halves that match as mirror images is being _____.

**2** A copy of a figure flipped across a line is its _____ _____.

**3** Reflective symmetry in plants and animals is called _____ _____.

**4** The line you draw so each side reflects the other is the _____ _____ _____.

**5** An even arrangement where neither side feels heavier is _____.

## Write About the Math

### How do you respond to your friend?

***¿Cómo le respondes a tu amigo?***

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **symmetric**
  *simétrico*
- ☐ **mirror image**
  *imagen reflejada*
- ☐ **bilateral symmetry**
  *simetría bilateral*
- ☐ **line of symmetry**
  *eje de simetría*
- ☐ **balance**
  *equilibrio*

**Model:** *A figure is symmetric only if there is a specific line where one side is a mirror image of the other.* — A strong answer explains the line gives you something to compare the two sides across — without it, you can't test whether the sides actually match. A weak answer just says 'it looks the same.'

### Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **To settle the disagreement, draw a ____ through the center of each flower and check whether one side is a ____ of the other. If the test works for both, then ____ flowers have bilateral symmetry.**
- **My answer is ____ because ____.**
  *Mi respuesta es ____ porque ____.*

### Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is ____.**
  *Mi afirmación es ____.*
- **My evidence is ____, so ____.**
  *Mi evidencia es ____, entonces ____.*

## Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

## Answer Key & Teacher Guide

1. **Notes 1:** symmetric
2. **Notes 2:** mirror image
3. **Notes 3:** bilateral symmetry
4. **Notes 4:** line of symmetry
5. **Notes 5:** balance
6. **Watch for this mistake:** *Thinking a figure lacks bilateral symmetry just because its line of symmetry is not vertical — the mirror line can run in any direction, and the test is whether the two sides match across it.*
7. **Try It 1:** C. 46 flowers — *One side uses 14 + 9 = 23 flowers, and the mirror side uses the same 23, so the whole design uses 23 × 2 = 46 flowers.*
8. **Try It 2:**